NullState: a reference-anchored framework for detecting, diagnosing, and quantifying the off-target compartment of human neural organoids

Luiz Takahashi[1]

[1] *Voiland College of Engineering and Architecture, Washington State University, Pullman, WA, USA , Correspondence:* `l.takahashidosreis@wsu.edu`

June 18, 2026

Abstract

Human neural organoids reproducibly generate cell types that lie outside the developmental trajectories they are intended to model, but the field lacks a principled, atlas-anchored way to *quantify* this off-target compartment and ask whether it carries reproducible structure. We present **NullState**, a reference-based pipeline that maps the 1,767,346-cell Human Neural Organoid Atlas (HNOCA) onto a harmonized 2,615,898-cell first-trimester human fetal reference (HDBCA + a fetal-body atlas) by variational reference surgery, assigns each organoid cell a pair of calibrated mapping scores (off-manifold distance and label entropy), and classifies it into a six-state off-target taxonomy. Across the atlas, 38.4% of cells fall into a large, *largely neural* ambiguous-novel (off-manifold) compartment; quantifying and explaining it is the work of Aim 2. Four orthogonal diagnostics show it is not a projection artifact but a *structured mixture* of a reference-coverage gap and a pervasive in-vitro stress program: off-manifold cells lose neural-progenitor identity while gaining hypoxia and unfolded-protein-response signatures, their burden rises monotonically with maturation and metabolic load, varies ∼240-fold across differentiation protocols (0.4% to 96.8%), and concentrates into coherent transcriptomic clusters; projecting these cells onto an *independent* prenatal cortex atlas confirms a genuine in-vitro-divergence component beyond the reference gap (ambiguous-novel cells sit further from that atlas than clean cells, Cliff's $\delta = 0.30$). Asking next whether lineage can be separated from the organoid-versus-primary shift, we report a reproducible *negative* result: a shared-latent contrastive model cannot. An origin-decoding adversary recovers organoid-versus-primary identity from the putatively dish-stripped latent at $\geq$0.998 balanced accuracy, robustly across latent-capacity and batch-conditioning interventions. Because the two sources occupy near-disjoint expression manifolds, this shift—biological in-vitro state and technical platform differences combined—is pervasive and encoder-visible, not a separable nuisance subspace; we therefore state the result methodologically (shared-latent methods cannot yield a source-invariant lineage axis here) and read the specifically in-vitro biology from the stress diagnostics rather than from the adversary. Finally, on the *non-neural* contaminant populations—where, unlike the reference-limited neural compartment, we are statistically powered—we show the method recovers real lineage geometry: all pairwise distances clear a matched-N noise floor (14–105×) and are biologically ordered, the two myogenic populations closest. Aim 3 thus demonstrates on the powered compartment what Aim 2's diagnosis implies for the neural one. NullState provides a deployable off-target quality axis for organoid studies and a precise account of when latent disentanglement is and is not attainable.

1 Introduction

Human pluripotent-stem-cell-derived neural organoids have become a central model of human brain development [1, 2], capturing cortical cytoarchitecture, diverse neuronal and glial identities, and functional network activity [3–5]. Yet a recurring concern limits their interpretation: organoids also produce cells that do not correspond to any *bona fide* in-vivo neural state. These include non-neural lineages arising from incomplete germ-layer restriction and aberrant states driven by the culture environment itself. A landmark observation is that metabolic and endoplasmic-reticulum (ER) stress in organoids impairs the specification of correct molecular subtypes [6], and cross-protocol comparisons against fetal tissue have repeatedly surfaced protocol-dependent, off-target differentiation routes [7, 8]. The recent Human Neural Organoid Atlas (HNOCA)—an integrated transcriptomic atlas of >1.7 million organoid cells spanning dozens of protocols [9]—makes it possible, for the first time, to study this off-target compartment at scale and against a common coordinate system.

What has been missing is a *principled, reference-anchored* way to (i) decide which organoid cells are off-target, (ii) explain *why* they are off-target, and (iii) ask whether the off-target compartment is merely noise or carries reproducible biological structure. Reference-based mapping provides the natural substrate: by projecting organoid cells onto a comprehensive fetal atlas, cells that map confidently to a fetal identity can be separated from those that do not. Two recently completed human references make such anchoring feasible—a comprehensive first-trimester human brain cell atlas (HDBCA) [10] and a human fetal-body gene-expression atlas [11]—together spanning neural and non-neural developmental origins.

Here we introduce **NullState**, a pipeline that operationalizes this idea (Fig. 1). We harmonize the two fetal atlases into a single 2,615,898-cell reference, map the 1,767,346-cell HNOCA query onto it by variational reference surgery [12–14], and assign every organoid cell two calibrated mapping scores—an off-manifold reconstruction distance and a label-assignment entropy—that jointly define a six-state off-target taxonomy. We organize the analysis around three questions: **(Aim 1)** is the off-target measurement well-posed and adequately powered? **(Aim 2)** what *is* the dominant, largely neural off-manifold compartment, and can lineage be separated from the in-vitro state so that those cells can be placed on a "dish-stripped" lineage axis? and **(Aim 3)** do off-target populations occupy distinct, reproducible geometry? The division of labor between the last two is deliberate and worth stating up front. Aim 2 quantifies and explains the neural off-manifold compartment, the bulk of the off-target signal but one that the present first-trimester reference cannot yet resolve into clean lineage geometry; Aim 3 then turns to the *non-neural* contaminant populations (skeletal and cardiac muscle, fibroblast, adrenal-cortical cells), the compartment where the reference *is* complete and the cell numbers *are* sufficient, and asks whether the method recovers real lineage structure there. Our central findings are that the off-manifold compartment is a structured biological mixture rather than an artifact; that shared-latent disentanglement provably fails here for an interpretable reason, so the negative is stated methodologically and the biology read from the diagnostics; and that off-target geometry, demonstrated on the powered non-neural compartment, is recoverable and biologically coherent.

2 Results

2.1 A harmonized fetal reference supports confident organoid mapping

We assembled a fetal reference by combining the first-trimester HDBCA [10] with the non-neural compartments of a human fetal-body atlas [11], retrieving raw unique-molecular-identifier (UMI) counts from the CZ CELLxGENE Discover Census [15]. The merged reference comprises 2,615,898 cells × 61,497 genes, with 32/33 lineage origins resolved. We trained a single-cell variational model (scVI) for 400 epochs (evidence lower bound, ELBO, 695.6), refined it into a label-aware model (scANVI, ELBO 736.1), and projected the HNOCA query onto the reference by architectural surgery (scArches; query-surgery ELBO 920.2) [12–14, 16]. Surgery required two stabilizers that we found essential at atlas scale: excluding 9,229 non-UMI (full-length read-count) cells whose library sizes destabilize the negative-binomial encoder, and filtering to cells with ≥ 50 counts across highly variable genes (1,767,346 → 1,689,762; 4.4% removed).

Donor-level batch integration improved cross-batch mixing while preserving biological structure: the integration local inverse Simpson's index (iLISI) rose from 6.10 (unintegrated principal components) to 8.10 after scVI, with only a modest change in the cell-type cLISI (1.31 → 1.44) [17, 18] (Fig. 2c). We calibrated two decision thresholds on held-out reference cells—a label-entropy threshold $\tau_H = 0.96$ and an off-manifold threshold $\tau_R = 1.93$—and used them to assign each organoid cell to one of six states (Fig. 2a,b). The resulting partition is dominated by an *ambiguous-novel* (off-manifold) compartment (677,783 cells; 38.4%), followed by clean on-target (459,441; 26.0%), poorly differentiated on-target (281,217; 15.9%), ambiguous (167,838; 9.5%), true off-target (93,795; 5.3%), and quality-control-dropped cells (86,813; 4.9%).

Three feasibility gates confirmed the measurement is well-posed (Fig. 2d). The off-target taxonomy is internally consistent (Gate A, pass); an in-vitro "dish-vector" computed on clean on-target cells is highly self-consistent (Gate B, mean cosine $0.855 > 0.70$), justifying a single shared in-vitro axis; and the true off-target compartment is adequately powered for downstream geometry (Gate C, green; 93,795 cells). NullState thus yields a calibrated, adequately powered off-target measurement over the entire organoid atlas.

2.2 The off-manifold compartment is a structured mixture of reference gap and in-vitro stress

The size of the ambiguous-novel compartment (38.4%) raises an essential question: is it a mapping artifact, or does it reflect real biology? Four orthogonal diagnostics, computed on the neural subset, converge on the latter (Fig. 3).

Identity and stress programs (Fig. 3a). Relative to clean on-target neural cells, ambiguous-novel neural cells show markedly reduced neural-progenitor identity (NPC score 1.11 → 0.61) while gaining canonical stress signatures: hypoxia (0.60 → 0.96) [19] and the unfolded-protein/ER-stress response (0.12 → 0.56, a ∼5-fold increase). An independent, atlas-precomputed glycolysis hallmark score is correspondingly elevated (0.46 → 0.65). This is the transcriptional signature of culture stress described by Bhaduri *et al.* [6], here resolved at atlas scale and tied directly to off-manifold status.

Maturation and metabolic load (Fig. 3b). The off-manifold fraction rises monotonically with a per-cell potency/complexity proxy (number of detected genes; quintile fractions 0.13 → 0.78) and is highest in the oldest organoids (age quintile fraction 0.62; median off-manifold score 1.84 → 2.12 across age) [20]. Off-target burden therefore accrues with culture time and metabolic demand, consistent with a stress-driven mechanism rather than random misassignment.

Protocol dependence (Fig. 3c). Across 26 differentiation protocols, the ambiguous-novel fraction spans nearly the entire range (0.4% to 96.8%; standard deviation 0.28). Guided cortical protocols sit at the low end (e.g., DAPT-patterned cortico-motor [21], 0.4%; 2, 10%), whereas regionally divergent or barrier-forming protocols dominate the high end—choroid-plexus organoids [22] reach 96.8%, midbrain-dopaminergic [23] and thalamic [24] protocols 85–90%. The off-manifold signal thus partly tracks *reference coverage*: the first-trimester, cortex-rich HDBCA under-represents later and non-cortical neural states, which are then read as off-manifold.

Cluster coherence (Fig. 3d,e). Ambiguous-novel cells are not scattered. Across 30 Leiden communities in the atlas embedding [25], the off-manifold fraction is bimodal (per-cluster s.d. 0.29 about an overall mean of 0.55); 20% of cells reside in near-pure off-manifold clusters ($\geq 80\%$) and 21% in near-pure clean clusters ($\leq 20\%$). Off-manifold status is a coherent population-level property, visible as contiguous territory in the scPoli UMAP [26].

Direct cross-atlas test (Fig. 6). The protocol gradient argues for a coverage-gap contribution indirectly; we tested it directly by asking whether off-manifold neural cells map onto an *independent*, non-HDBCA reference—the prenatal human cortex of Velmeshev *et al.* [27]—reasoning that cells merely missing from HDBCA should be recoverable there, whereas genuinely divergent cells should not. The experiment exposes, and is bounded by, the same pervasiveness that defeats Aim 2: the organoid-versus-primary source gap displaces *every* organoid cell from the independent atlas, including clean on-target cells that map confidently to HDBCA (clean median off-manifold distance 4.0 versus reference-self 2.2), under both scArches projection and joint scVI integration. Absolute "mapped-on" fractions are therefore confounded, and we do not report them. Instead the clean control *quantifies* that shared displacement, and—exactly as in the Aim 3 cancellation argument—the clean-versus-ambiguous-novel contrast removes it: ambiguous-novel neural cells are significantly and consistently further from the independent cortex manifold than clean cells (area under the curve 0.65, Cliff's $\delta = 0.30$, $p < 10^{-300}$; 70% exceed the clean median and 15% lie beyond the clean 95th percentile). This is direct, confound-controlled evidence for a genuine in-vitro-divergence component in the off-manifold compartment, over and above the coverage gap implied by the protocol gradient.

Together these diagnostics establish that the dominant off-target compartment is a reproducible biological mixture: a *reference-coverage gap* (non-cortical and later neural states absent from a first-trimester reference) superimposed on a pervasive *in-vitro stress* program (hypoxia, unfolded-protein response, glycolysis) that intensifies with maturation.

2.3 Lineage cannot be disentangled from in-vitro state in a shared latent (a characterized negative)

If the in-vitro state were a low-dimensional nuisance, one could subtract it and place off-target cells on a clean lineage axis. We tested this directly with contrastiveVI, which factorizes a background (shared) latent z_{lin} from a salient latent z_{iv} [28], training the background on primary fetal cells and the salient on organoid cells, so that z_{lin} should encode dish-stripped lineage. We evaluated disentanglement with a held-out adversary: a classifier that attempts to recover organoid-versus-primary origin from z_{lin} (balanced accuracy 0.5 is ideal; < 0.6 would indicate success) (Fig. 4a).

Disentanglement failed, and did so robustly. The baseline model ($\dim z_{\text{iv}} = 10$) yielded adversary balanced accuracy 0.998; tripling salient capacity ($\dim z_{\text{iv}} = 30$) gave 0.999; and adding a source batch covariate to the decoder—the textbook remedy for a nuisance axis confounded with a condition—gave 0.998. Training converged normally in every case (Fig. 4b), so this is not an optimization failure. The mechanism is structural and, given how different the two sources are, largely *expected*: the contribution here is to characterize precisely *why* a shared latent cannot work, not to present the failure as a surprise. When organoid and fetal cells occupy near-disjoint regions of expression space, source identity is almost definitionally decodable and there is no overlap region in which a shared lineage axis is even well defined—the background *encoder* $q(z_{\text{lin}} \,|\, x)$ observes and encodes the organoid-versus-primary difference directly, and conditioning the *decoder* on a batch label cannot undo what the encoder has already written into z_{lin} (Fig. 4c). Two scope limits on the interpretation follow, and we are explicit about them. First, the adversary establishes only that this difference is decodable, not what it is made of: organoid and primary datasets differ in biological in-vitro state *and* in laboratory, dissociation, platform, and ambient-RNA profile, every one of which is transcriptome-pervasive and would on its own render source decodable. The experiment therefore supports a claim we can fully defend—shared-latent contrastive methods cannot produce a source-invariant lineage axis in this regime—whereas the specifically *in-vitro* character of the dominant axis is established not by the adversary but by the stress diagnostics of the previous section (Fig. 3). Second, the donor-keyed integration used for mapping does not correct the far deeper organoid-versus-primary divide, so that divide is precisely what the background latent inherits. We accordingly state the negative result methodologically and read its biology from Aim 2's signatures. This delimits where "calibration without anchoring" can and cannot be applied and motivates encoder-side or distributional alternatives.

2.4 Off-target populations occupy distinct, biologically coherent geometry

Having quantified the largely neural off-manifold compartment in Aim 2, we turn to the part of the off-target landscape where the reference is complete and the populations are large enough to support geometry: the *non-neural* contaminant lineages. This division of labor is deliberate. The neural off-manifold cells are the larger compartment, but, as Aim 2 showed, a first-trimester reference cannot yet resolve them into clean lineage structure (it conflates genuine in-vitro divergence with its own coverage gaps); the non-neural contaminants—skeletal and cardiac muscle, fibroblast, and adrenal-cortical cells—instead map to a *complete* fetal-body reference and number in the thousands, making them the compartment on which the geometry method can be put to a fair test. The disentanglement failure does not invalidate the geometry of these populations relative to one another. Aim 3 compares off-target *organoid* populations against *each other*; because all such cells share the same organoid-versus-primary offset in z_{lin}, that offset is a constant translation that cancels exactly in pairwise sliced-Wasserstein distances ($W(\mathcal{A}+c, \mathcal{B}+c) = W(\mathcal{A}, \mathcal{B})$) [29]. The contamination shifts all populations together and therefore does not distort their relative structure.

We computed pairwise lineage distances among the true off-target populations, controlling for sample size by a rarefaction-derived minimum ($N_{\min} = 2000$) and a matched-N self-distance noise floor; a pair is called only if its cross-distance confidence interval clears the floor (Fig. 5). Of the 17 off-target populations, four *non-neural contaminant* lineages are well powered (cortical-adrenal 49,338; skeletal-muscle 30,629; fibroblast 6,349; cardiac-muscle 2,534). All six pairwise comparisons are real differences, each clearing the noise floor by 14 to 105× (W from 0.32 to 0.99 against floors of 0.007–0.026); the remaining 130 small-N comparisons are correctly returned as underpowered. Crucially, the recovered geometry is *biologically ordered*: the two myogenic populations (cardiac and skeletal muscle) are closest ($W = 0.32$), fibroblast is most distant from the muscle populations ($W \approx 0.96$–0.99), and the steroidogenic adrenal-cortical population sits between them, exactly as a lineage embedding—not a source-confounded one—should arrange them (Fig. 5a,d). Off-target geometry is thus both statistically real and lineage-interpretable.

3 Discussion

NullState turns the off-target problem of neural organoids into a quantitative, atlas-anchored measurement. Three results have practical consequences. First, the off-manifold compartment is large (38.4%) but *legible*: it decomposes into a reference-coverage gap and an in-vitro stress program, each separately actionable. The reference-gap component argues for extending fetal references beyond the first trimester and beyond cortex before off-manifold scores are interpreted as "aberrant" for non-cortical protocols; the stress component, intensifying with age and metabolic load and mirroring Bhaduri et al. [6], provides a per-cell, deployable readout for evaluating stress-mitigation strategies. The ∼240-fold protocol spread additionally yields a concrete, reference-anchored fidelity ranking across organoid platforms.

Second, our negative disentanglement result is useful precisely because it is characterized, and we are deliberate about what it does and does not claim. The widespread intuition that a contrastive or batch-aware latent can "subtract the dish" presupposes that the organoid-versus-primary difference is a separable nuisance subspace. We show that in this regime it is not—it is a pervasive, encoder-visible shift—so shared-latent methods cannot produce a source-invariant lineage axis regardless of salient capacity or decoder batch conditioning. That methodological conclusion holds whether the shift is biological or technical in origin; the separate evidence that its dominant component is specifically the in-vitro stress state comes from the Aim 2 diagnostics, not from the adversary, and the two should not be conflated. Methods that intervene at the encoder (e.g., adversarial or invariance-penalized encoders) or that operate distributionally rather than per-cell are the appropriate next step.

Third, and reassuringly, the quantity Aim 3 actually needs—the *relative* geometry of off-target populations—survives the disentanglement failure by an exact cancellation argument, and recovers biologically sensible lineage structure. This means off-target geometry can be used as a quality and comparison axis today, even though a fully dish-stripped per-cell coordinate is not yet attainable.

Limitations follow directly from the analysis. Off-manifold status conflates reference gap with biological aberrance, so absolute off-target rates are reference-dependent and should be read comparatively; indeed the pervasive organoid-versus-primary source gap displaces even clean, HDBCA-validated cells from an independent atlas, so the *absolute* coverage-gap fraction is not directly measurable (Fig. 6). The protocol gradient and the clean-calibrated cross-atlas contrast are the available evidence, and they agree on a mixture rather than on either pole. The dish-vector and stress scores are signature-based proxies, partially corroborated by the maturation correlation and the independent-atlas divergence signal. The well-powered Aim 3 populations are non-neural off-target lineages; neural off-target geometry awaits a reference that closes the coverage gap. Finally, pairwise distances are robust to the global source offset but may still include a component of *differential* in-vitro state between populations; the lineage-consistent ordering (myogenic populations closest) indicates that lineage dominates, but isolating that component is future work.

4 Methods

Reference construction. Raw-count fetal data were retrieved from the CZ CELLxGENE Discover Census [15]: the full first-trimester HDBCA [10] as the neural core, and non-neural compartments of a fetal-body atlas [11] filtered at the Census level to the relevant ontology cell types (mesoderm, endoderm, neural-crest origins). Gene identifiers were harmonized to Ensembl; counts were verified as integer UMIs prior to training. A lineage *origin* label was assigned to each reference cell through a curated cell-type→origin map; only the generic "cell" type was permitted to remain unmapped.

Mapping by reference surgery. We trained scVI (400 epochs, donor batch key) and scANVI on the reference, then mapped the HNOCA query [9] by scArches architectural surgery, using the implementations in scvi-tools [12–14, 16]. Organoid raw UMI counts were taken from the CELLxGENE HNOCA copy. Surgery stability at atlas scale required (i) dropping non-UMI read-count cells (per-cell maximum count > 1000) and (ii) a learning-rate, gradient-clipping, and KL-warmup schedule; without these the negative-binomial encoder diverged to NaN.

Scoring and classification. For each query cell we computed a label-assignment entropy H over scANVI posteriors and an off-manifold reconstruction distance ρ in the latent space. Thresholds τ_H and τ_R were calibrated on a held-out 10% of reference cells (off-manifold nearest-neighbor distances fit excluding the held-out set to avoid self-neighbor deflation). The six-state taxonomy combines on/off-manifold status (ρ vs. τ_R), label confidence (H vs. τ_H), and the lineage origin of the assigned label.

Diagnostics. Gene-set scores (neural progenitor, neuron, glia, hypoxia [19], unfolded-protein response, glycolysis) were computed with `scanpy.tl.score_genes` on log-normalized expression. Maturation was assessed against organoid age and a detected-gene potency proxy [20]; protocol stratification used the atlas `assay_differentiation` annotation; cluster coherence used Leiden communities [25] on the scPoli embedding.

Disentanglement. contrastiveVI [28] was trained with primary fetal on-target cells as background and organoid on-target cells as target (100,000 each, 2000 highly variable genes, raw-count layer). Disentanglement was scored by a five-fold cross-validated logistic-regression adversary predicting organoid-versus-primary origin from the background latent z_{lin} (balanced accuracy). Interventions: salient dimension $10 \to 30$, and a source batch covariate registered through `setup_anndata`.

Second-reference test. To probe coverage gap versus divergence directly, we jointly integrated 30,000 ambiguous-novel and 15,000 clean-on-target neural organoid cells with 100,000 prenatal cells of an independent cortical atlas (Velmeshev *et al.* [27]; gestational "LMP-month" stages, drawn from the CELLxGENE Census) using scVI with the dataset as batch key (3000 shared highly variable genes, 30-dimensional latent). Each organoid cell's off-manifold score is its mean distance to the 15 nearest reference cells in the integrated latent. Because clean and ambiguous-novel cells share the same organoid-versus-reference displacement, the absolute on-manifold fraction is confounded; we therefore calibrate against the clean distribution and report the clean-versus-ambiguous-novel effect size (Mann–Whitney AUC, Cliff's δ) and the fraction of ambiguous-novel cells beyond clean percentiles.

Geometry. For each off-target population pair we estimated the sliced-Wasserstein distance [29] between background-latent embeddings, with a per-population minimum sample size from rarefaction ($N_{\text{min}} = 2000$) and a matched-N self-distance noise floor (95th percentile). A pair was called a real difference when its bootstrap cross-distance interval exceeded the floor; multiplicity was controlled by Benjamini–Hochberg [30]. Sliced-Wasserstein pairwise distances are invariant to a translation shared by both populations, which is what makes the comparison valid despite the residual source offset in z_{lin}.

Compute and code. Models were trained on a single NVIDIA H100 GPU. All quantitative panels are generated directly from pipeline artifacts (`docs/manuscript/make_figures.py`); numerical values are exported to `numbers.json`.

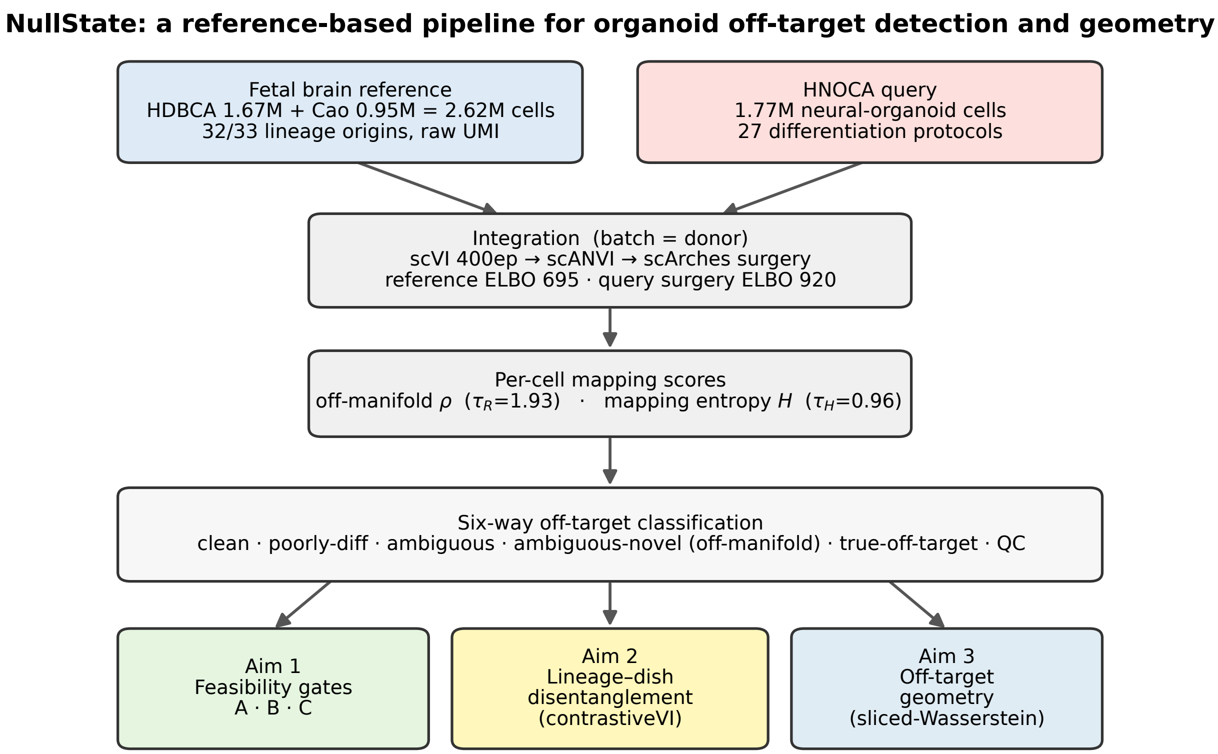

Figure 1. The NullState pipeline. A harmonized first-trimester fetal reference (HDBCA + fetal-body atlas; 2,615,898 cells, raw UMIs) and the HNOCA organoid query (1,767,346 cells, 27 protocols) are co-embedded by scVI/scANVI followed by scArches reference surgery. Each organoid cell receives two calibrated mapping scores—off-manifold distance ρ ($\tau_R = 1.93$) and label entropy H ($\tau_H = 0.96$)—that define a six-state off-target taxonomy, which feeds three analyses: feasibility gates (Aim 1), lineage–dish disentanglement (Aim 2), and off-target geometry (Aim 3).

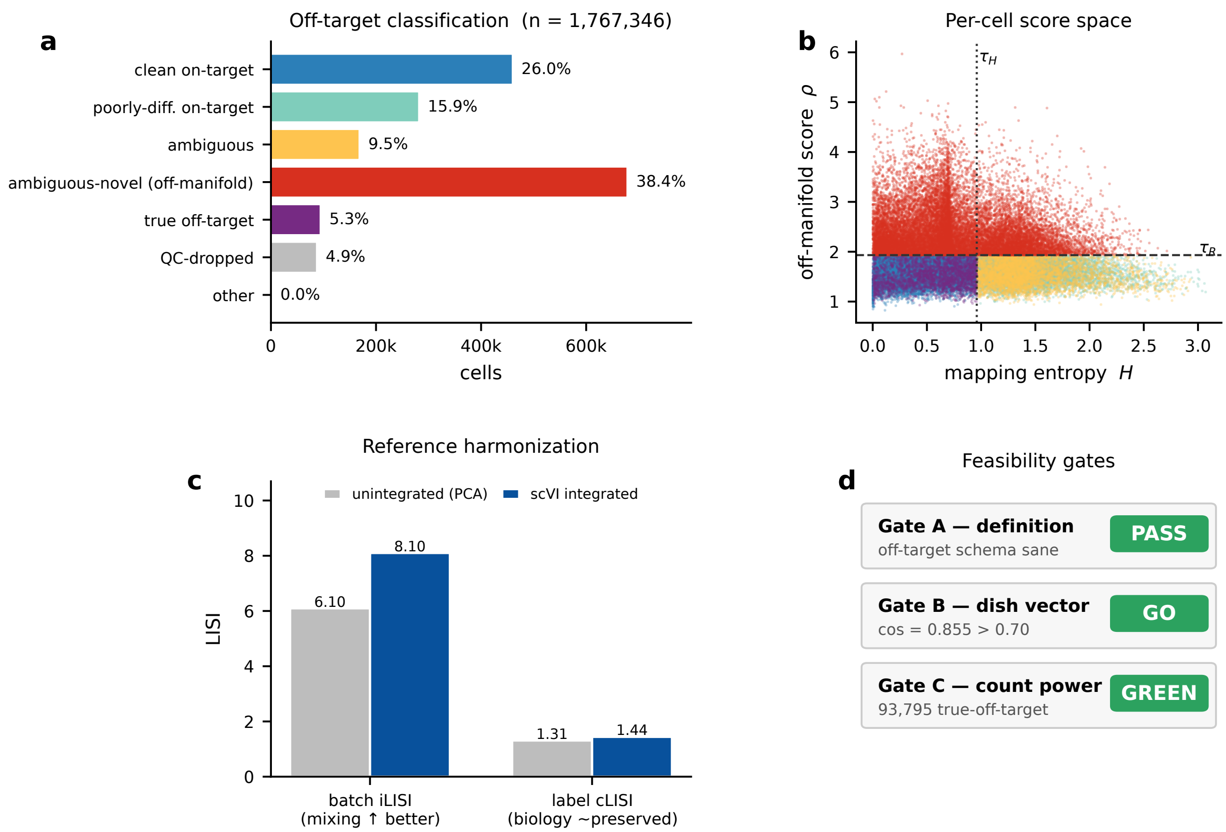

Figure 2. Harmonized mapping and the off-target landscape (Aim 1). (**a**) Six-state classification of all 1,767,346 organoid cells; the off-manifold *ambiguous-novel* state dominates (38.4%). (**b**) Per-cell score space: mapping entropy H versus off-manifold score ρ (subsample), coloured by class, with the calibrated thresholds τ_H (dotted) and τ_R (dashed). (**c**) Donor-batch integration improves mixing (iLISI $6.10 \to 8.10$) while preserving biology (cLISI $1.31 \to 1.44$). (**d**) The three feasibility gates pass: definition (A), dish-vector self-consistency (B, cosine 0.855), and off-target power (C, green).

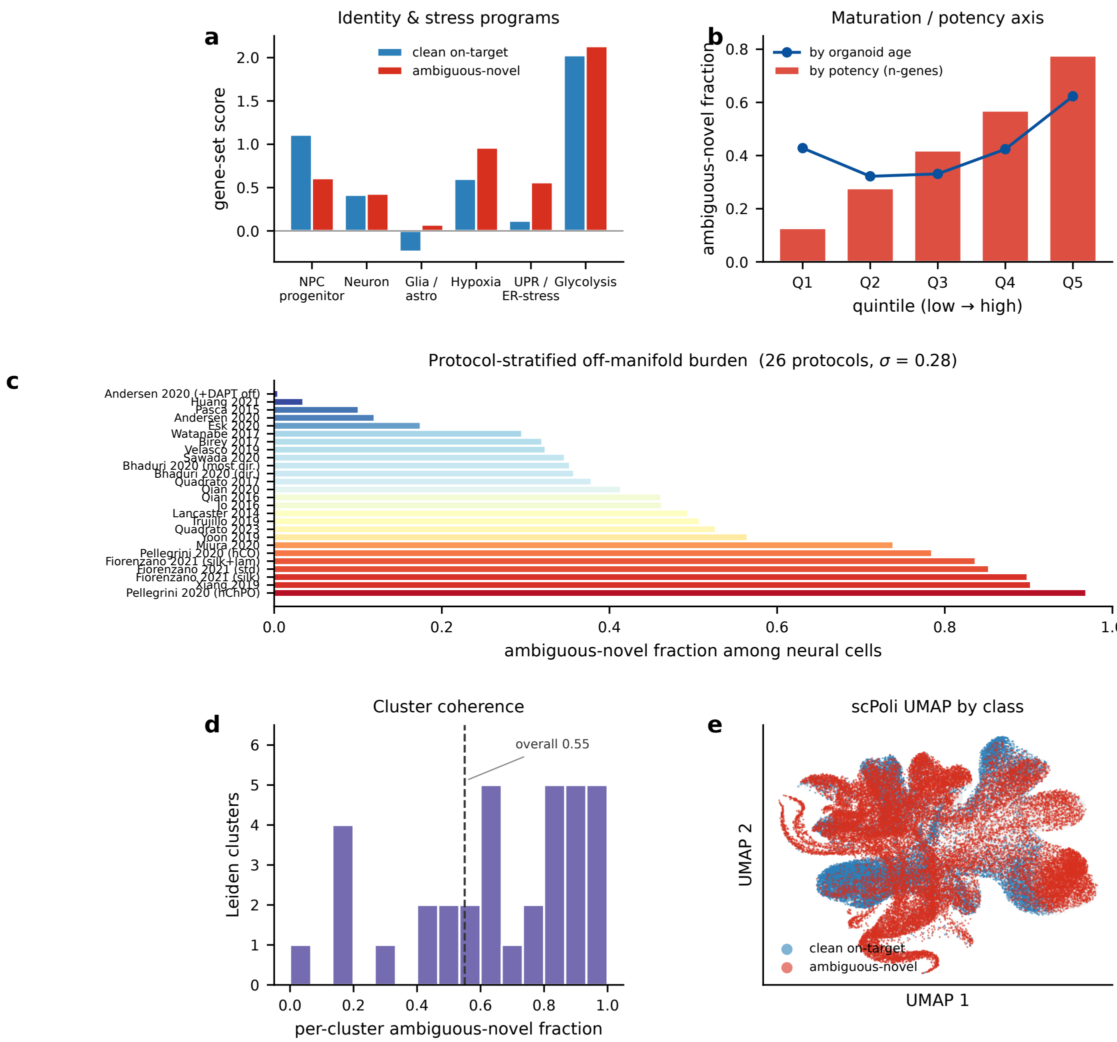

Figure 3. The off-manifold compartment is structured, not artifactual. (**a**) Ambiguous-novel neural cells lose neural-progenitor identity and gain hypoxia and unfolded-protein/ER-stress signatures relative to clean on-target neural cells. (**b**) Off-manifold fraction increases monotonically with a detected-gene potency proxy (bars) and with organoid age (line). (**c**) Protocol-stratified off-manifold burden spans 0.4%–96.8% across 26 protocols (colour = fraction); guided cortical protocols are lowest, choroid-plexus/midbrain/thalamic highest. (**d**) Per-cluster off-manifold fraction across 30 Leiden communities is bimodal (coherent populations, not scatter). (**e**) scPoli UMAP coloured by class: ambiguous-novel cells occupy contiguous territory.

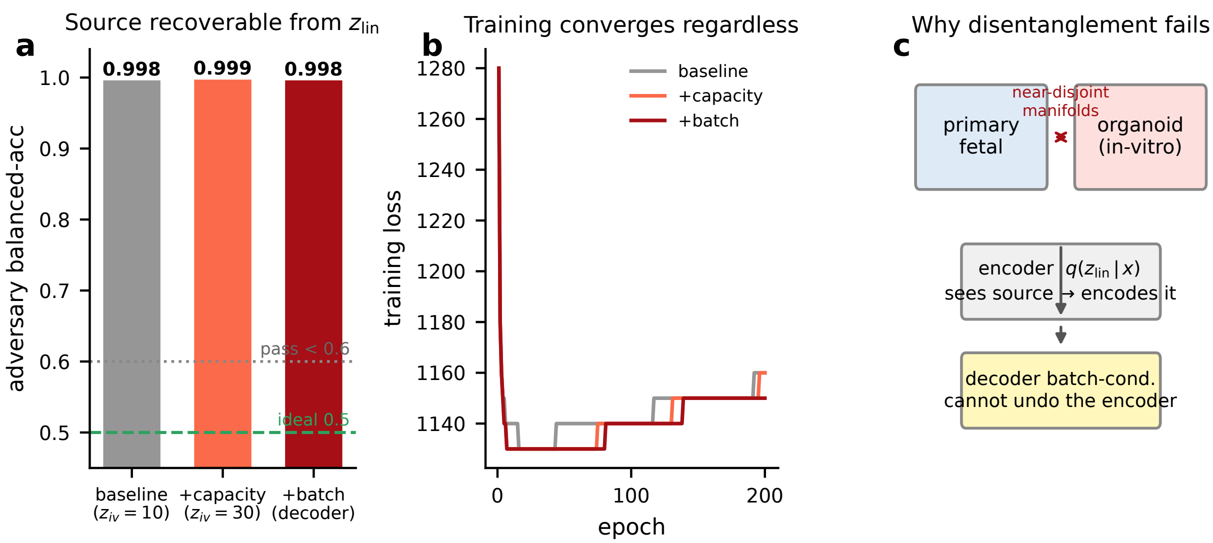

Figure 4. Lineage–dish disentanglement is not attainable (characterized negative; Aim 2). (**a**) An origin-decoding adversary recovers organoid-versus-primary identity from the "dish-stripped" latent z_{lin} at ≥ 0.998 balanced accuracy across all interventions (ideal 0.5; pass < 0.6). (**b**) Training converges normally in every run, so the failure is structural, not optimization-related. (**c**) Mechanism: near-disjoint primary/organoid manifolds make the background *encoder* encode source directly; decoder batch-conditioning cannot remove it.

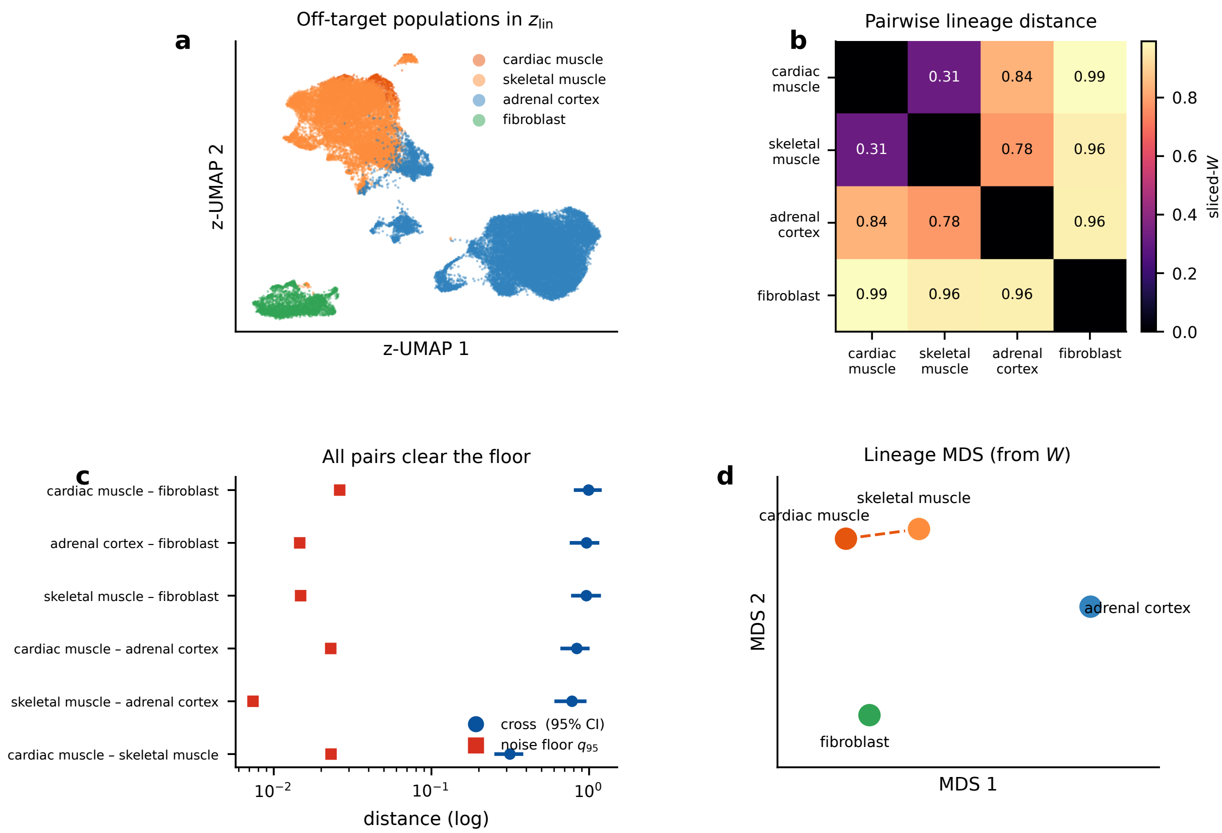

Figure 5. Off-target geometry is real and biologically ordered (Aim 3). (**a**) Background-latent (z_{lin}) embedding of the off-target populations; the four well-powered lineages separate. (**b**) Pairwise sliced-Wasserstein lineage-distance matrix. (**c**) Every well-powered pair clears its matched-N noise floor (q_{95}, red) by 14–105× (cross-distance with 95% CI, blue; log axis). (**d**) Multidimensional scaling of the distance matrix: the two myogenic populations (cardiac, skeletal muscle) are closest, fibroblast most distant, as expected for a lineage embedding.

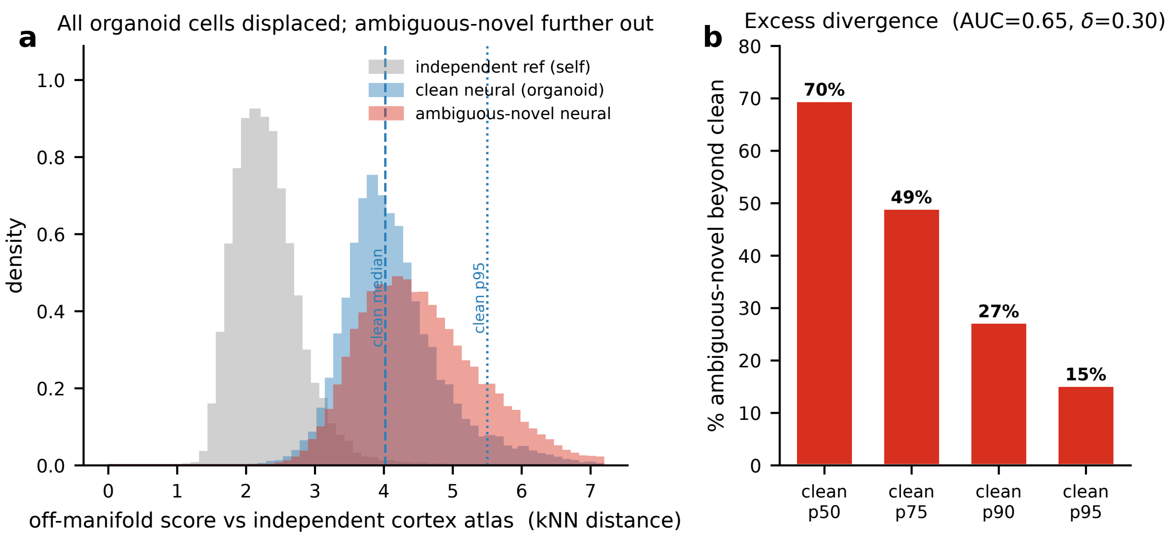

Figure 6. Direct cross-atlas test of coverage gap versus divergence. Ambiguous-novel and clean-on-target neural organoid cells were jointly integrated with an *independent* prenatal cortical atlas (Velmeshev *et al.* 2023 [27]). (**a**) Off-manifold score (kNN distance to the independent reference) for reference-self (grey), clean neural (blue), and ambiguous-novel neural (red). The organoid-versus-reference source gap displaces *all* organoid cells—including clean cells—far from the reference (clean/ambiguous medians 4.0/4.4 versus reference-self 2.2), so absolute mapped-on fractions are confounded; the clean distribution (dashed median, dotted 95th percentile) is the calibrator. (**b**) The clean-versus-ambiguous contrast cancels the shared displacement: 70% of ambiguous-novel cells exceed the clean median and 15% lie beyond the clean 95th percentile (overall AUC $= 0.65$, Cliff's $\delta = 0.30$), isolating a genuine in-vitro-divergence component.

References

[1] Madeline A. Lancaster, Magdalena Renner, Carol-Anne Martin, Daniel Wenzel, Louise S. Bicknell, Matthew E. Hurles, Tessa Homfray, Josef M. Penninger, Andrew P. Jackson, and Jürgen A. Knoblich. Cerebral organoids model human brain development and microcephaly. *Nature*, 501 (7467):373–379, 2013. doi: 10.1038/nature12517.

[2] Anca M. Paşca, Steven A. Sloan, Laura E. Clarke, Yuan Tian, Christopher D. Makinson, Nina Huber, Chul Hoon Kim, Jin-Young Park, Nancy A. O'Rourke, Khoa D. Nguyen, Stephen J. Smith, John R. Huguenard, Daniel H. Geschwind, Ben A. Barres, and Sergiu P. Paşca. Functional cortical neurons and astrocytes from human pluripotent stem cells in 3d culture. *Nature Methods*, 12(7):671–678, 2015. doi: 10.1038/nmeth.3415.

[3] Giorgia Quadrato, Tuan Nguyen, Evan Z. Macosko, John L. Sherwood, Sung Min Yang, Daniel R. Berger, Natalie Maria, Jorg Scholvin, Melissa Goldman, Justin P. Kinney, Edward S. Boyden, Jeff W. Lichtman, Ziv M. Williams, Steven A. McCarroll, and Paola Arlotta. Cell diversity and network dynamics in photosensitive human brain organoids. *Nature*, 545(7652): 48–53, 2017. doi: 10.1038/nature22047.

[4] Silvia Velasco, Amanda J. Kedaigle, Sean K. Simmons, Allison Nash, Marina Rocha, Giorgia Quadrato, Bruna Paulsen, Lan Nguyen, Xian Adiconis, Aviv Regev, Joshua Z. Levin, and Paola Arlotta. Individual brain organoids reproducibly form cell diversity of the human cerebral cortex. *Nature*, 570(7762):523–527, 2019. doi: 10.1038/s41586-019-1289-x.

[5] Cleber A. Trujillo, Richard Gao, Priscilla D. Negraes, Jing Gu, Justin Buchanan, Sebastian Preissl, Allen Wang, Wei Wu, Gabriel G. Haddad, Isaac A. Chaim, Alain Domissy, Matthieu Vandenberghe, Anna Devor, Gene W. Yeo, Bradley Voytek, and Alysson R. Muotri. Complex oscillatory waves emerging from cortical organoids model early human brain network development. *Cell Stem Cell*, 25(4):558–569.e7, 2019. doi: 10.1016/j.stem.2019.08.002.

[6] Aparna Bhaduri, Madeline G. Andrews, Walter Mancia Leon, Diane Jung, David Shin, Denise Allen, Dana Jung, Galina Schmunk, Maximilian Haeussler, Jahan Salma, Alex A. Pollen, Tomasz J. Nowakowski, and Arnold R. Kriegstein. Cell stress in cortical organoids impairs molecular subtype specification. *Nature*, 578(7793):142–148, 2020. doi: 10.1038/s41586-020-1962-0.

[7] Yoshiaki Tanaka, Bilal Cakir, Yangfei Xiang, Gareth J. Sullivan, and In-Hyun Park. Synthetic analyses of single-cell transcriptomes from multiple brain organoids and fetal brain. *Cell Reports*, 30(6):1682–1689.e3, 2020. doi: 10.1016/j.celrep.2020.01.038.

[8] Sabina Kanton, Michael James Boyle, Zhisong He, Malgorzata Santel, Anne Weigert, Fátima Sanchís-Calleja, Patricia Guijarro, Leila Sidow, Jonas Simon Fleck, Dingding Han, Zhengzong Qian, Michael Heide, Wieland B. Huttner, Philipp Khaitovich, Svante Pääbo, Barbara Treutlein, and J. Gray Camp. Organoid single-cell genomic atlas uncovers human-specific features of brain development. *Nature*, 574(7778):418–422, 2019. doi: 10.1038/s41586-019-1654-9.

[9] Zhisong He, Leander Dony, Jonas Simon Fleck, Leila Sidow, Sabina Kanton, Barbara Treutlein, Fabian J. Theis, and J. Gray Camp. An integrated transcriptomic cell atlas of human neural organoids. *Nature*, 635(8039):690–698, 2024. doi: 10.1038/s41586-024-08172-8.

[10] Emelie Braun, Miri Danan-Gotthold, Lars E. Borm, Kyung Won Lee, Elin Vinsland, Peter Lönnerberg, Lijuan Hu, Xiaofei Li, Xiaoling He, Žaneta Andrusivová, Joakim Lundeberg, Ernest Arenas, Roger A. Barker, Erik Sundström, and Sten Linnarsson. Comprehensive cell atlas of the first-trimester developing human brain. *Science*, 382(6667):eadf1226, 2023. doi: 10.1126/science.adf1226.

[11] Junyue Cao, Diana R. O'Day, Hannah A. Pliner, Paul D. Kingsley, Mei Deng, Riza M. Daza, Michael A. Zager, Kimberly A. Aldinger, Ronnie Blecher-Gonen, Fan Zhang, Malte Spielmann, James Palis, Dan Doherty, Frank J. Steemers, Ian A. Glass, Cole Trapnell, and Jay Shendure. A human cell atlas of fetal gene expression. *Science*, 370(6518):eaba7721, 2020. doi: 10.1126/science.aba7721.

[12] Romain Lopez, Jeffrey Regier, Michael B. Cole, Michael I. Jordan, and Nir Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15(12):1053–1058, 2018. doi: 10.1038/s41592-018-0229-2.

[13] Chenling Xu, Romain Lopez, Edouard Mehlman, Jeffrey Regier, Michael I. Jordan, and Nir Yosef. Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models. *Molecular Systems Biology*, 17(1):e9620, 2021. doi: 10.15252/msb.20209620.

[14] Mohammad Lotfollahi, Mohsen Naghipourfar, Malte D. Luecken, Matin Khajavi, Maren Büttner, Marco Wagenstetter, Žiga Avsec, Adam Gayoso, Nir Yosef, Marta Interlandi, Sergei Rybakov, Alexander V. Misharin, and Fabian J. Theis. Mapping single-cell data to reference atlases by transfer learning. *Nature Biotechnology*, 40(1):121–130, 2022. doi: 10.1038/s41587-021-01001-7.

[15] CZI Cell Science Program. CZ CELLxGENE Discover: a single-cell data platform for scalable exploration, analysis and modeling of aggregated data. *Nucleic Acids Research*, 53(D1):D886–D900, 2025. doi: 10.1093/nar/gkae1142.

[16] Adam Gayoso, Romain Lopez, Galen Xing, Pierre Boyeau, Valeh Valiollah Pour Amiri, Justin Hong, Katherine Wu, Michael Jayasuriya, Edouard Mehlman, Maxime Langevin, Yining Liu, Jules Samaran, Gabriel Misrachi, Achille Nazaret, Oscar Clivio, Chenling Xu, Tal Ashuach, Mariano Gabitto, Mohammad Lotfollahi, Valentine Svensson, Eduardo da Veiga Beltrame, Vitalii Kleshchevnikov, Carlos Talavera-López, Lior Pachter, Fabian J. Theis, Aaron Streets, Michael I. Jordan, Jeffrey Regier, and Nir Yosef. A Python library for probabilistic analysis of single-cell omics data. *Nature Biotechnology*, 40(2):163–166, 2022. doi: 10.1038/s41587-021-01206-w.

[17] Ilya Korsunsky, Nghia Millard, Jean Fan, Kamil Slowikowski, Fan Zhang, Kevin Wei, Yuriy Baglaenko, Michael Brenner, Po-Ru Loh, and Soumya Raychaudhuri. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, 16(12):1289–1296, 2019. doi: 10.1038/s41592-019-0619-0.

[18] Malte D. Luecken, Maren Büttner, Kridsadakorn Chaichoompu, Anna Danese, Marta Interlandi, Michaela F. Müller, Daniel C. Strobl, Luke Zappia, Martin Dugas, Maria Colomé-Tatché, and Fabian J. Theis. Benchmarking atlas-level data integration in single-cell genomics. *Nature Methods*, 19(1):41–50, 2022. doi: 10.1038/s41592-021-01336-8.

[19] Francesca M. Buffa, Adrian L. Harris, Catharine M. West, and Crispin J. Miller. Large meta-analysis of multiple cancers reveals a common, compact and highly prognostic hypoxia metagene. *British Journal of Cancer*, 102(2):428–435, 2010. doi: 10.1038/sj.bjc.6605450.

[20] Gunsagar S. Gulati, Shaheen S. Sikandar, Daniel J. Wesche, Anoop Manjunath, Anjan Bharadwaj, Mark J. Berger, Francisco Ilagan, Angera H. Kuo, Robert W. Hsieh, Shang Cai, Maider Zabala, Ferenc A. Scheeren, Neethan A. Lobo, Dalong Qian, Feiqiao B. Yu, Frederick M. Dirbas, Michael F. Clarke, and Aaron M. Newman. Single-cell transcriptional diversity is a hallmark of developmental potential. *Science*, 367(6476):405–411, 2020. doi: 10.1126/science.aax0249.

[21] Jimena Andersen, Omer Revah, Yuki Miura, Nicholas Thom, Neal D. Amin, Kevin W. Kelley, Mandeep Singh, Xiaoyu Chen, Mayuri V. Thete, Elisabeth M. Walczak, Hannes Vogel, H. Christina Fan, and Sergiu P. Paşca. Generation of functional human 3d cortico-motor assembloids. *Cell*, 183(7):1913–1929.e26, 2020. doi: 10.1016/j.cell.2020.11.017.

[22] Laura Pellegrini, Claudia Bonfio, Jessica Chadwick, Farida Begum, Mark Skehel, and Madeline A. Lancaster. Human CNS barrier-forming organoids with cerebrospinal fluid production. *Science*, 369(6500):eaaz5626, 2020. doi: 10.1126/science.aaz5626.

[23] Alessandro Fiorenzano, Edoardo Sozzi, Marcella Birtele, Janko Kajtez, Jessica Giacomoni, Fredrik Nilsson, Andreas Bruzelius, Yogita Sharma, Yu Zhang, Bengt Mattsson, Jenny Emnéus, Daniella Rylander Ottosson, Petter Storm, and Malin Parmar. Single-cell transcriptomics captures features of human midbrain development and dopamine neuron diversity in brain organoids. *Nature Communications*, 12(1):7302, 2021. doi: 10.1038/s41467-021-27464-5.

[24] Yangfei Xiang, Yoshiaki Tanaka, Bilal Cakir, Benjamin Patterson, Kun-Yong Kim, Pingnan Sun, Young-Jin Kang, Mei Zhong, Xinran Liu, Prabir Patra, Sang-Hun Lee, Sherman M. Weissman, and In-Hyun Park. hESC-derived thalamic organoids form reciprocal projections when fused with cortical organoids. *Cell Stem Cell*, 24(3):487–497.e7, 2019. doi: 10.1016/j.stem.2018.12.015.

[25] Vincent A. Traag, Ludo Waltman, and Nees Jan van Eck. From Louvain to Leiden: guaranteeing well-connected communities. *Scientific Reports*, 9(1):5233, 2019. doi: 10.1038/s41598-019-41695-z.

[26] Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018. doi: 10.48550/arXiv.1802.03426.

[27] Dmitry Velmeshev, Yonatan Perez, Zhuoran Yan, Jonathan E. Valencia, David R. Castaneda-Castellanos, Li Wang, Lucas Schirmer, Simone Mayer, Brittney Wick, Shaohui Wang, Tomasz J. Nowakowski, Mercedes F. Paredes, Eric J. Huang, and Arnold R. Kriegstein. Single-cell analysis of prenatal and postnatal human cortical development. *Science*, 382(6667): eadf0834, 2023. doi: 10.1126/science.adf0834.

[28] Ethan Weinberger, Chris Lin, and Su-In Lee. Isolating salient variations of interest in single-cell data with contrastivevi. *Nature Methods*, 20(9):1336–1345, 2023. doi: 10.1038/s41592-023-01955-3.

[29] Nicolas Bonneel, Julien Rabin, Gabriel Peyré, and Hanspeter Pfister. Sliced and Radon Wasserstein barycenters of measures. *Journal of Mathematical Imaging and Vision*, 51(1):22–45, 2015. doi: 10.1007/s10851-014-0506-3.

[30] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.